

2024 AMC 10A Problem 5 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 5. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 5

Problem:

What is the least value of n such that $n!$ is a multiple of 2024 ?

Answer Choices:

- A. 11
- B. 21
- C. 22
- D. 23
- E. 253

Solution:

Because the prime factorization of 2024 is $2^3 \cdot 11 \cdot 23$, it follows that $n!$ is a multiple of 2024 if and only if $n \geq 23$. Therefore (B) 23 is the least value of n such that $n!$ is a multiple of 2024 .

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) [↗](#)

